

ATLANTIC COMPUTATIONAL EXCELLENCE NETWORK

Free Cluster Computing For Researchers

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What is ACEnet?

- “Atlantic Computational Excellence Network”
- Shared regional resource for ...
 - ... high-performance computing
 - ... remote collaboration
 - ... visualization
- Hardware, software, and personnel

What's “High Performance Computing”?



Many ordinary CPUs working on the same problem

- Multiple single-CPU jobs can be run in parallel
- Or a single job can involve many CPUs, communicating with one another
- Objective:
Complete the research in the shortest amount of time
- How?
Computer clusters --- Racks of regular computers



Computer clusters

Processors: AMD Opterons, 2.4 GHz thru 3.0 GHz

Cluster	Total Cores	RAM/node	Cores/node	Inter-connect	Best use
Brasdor	1152	8 Gb	2, 4	GigE	serial
Courtenay	52	16 Gb	4	GigE	development
Fundy	632	16-128	4, 16	IB	shared mem
Glooscap	848	4 Gb	4, 16	GigE	any
Mahone	556	16-64	4	Myr	MPI
Placentia	2680 +400	16-64	4, 16	IB, GigE	any

GigE = Gigabit Ethernet

IB = InfiniBand

Myr = Myrinet

https://wiki.ace-net.ca/index.php/Compute_Overview

CPUs

Cluster	Nodes	Cores/node	RAM/node	Total cores
Brasdor			8G	1152
Courtenay	13	4	16G	52
Fundy				
Glooscap				
Mahone				
Placentia				

Storage

Cluster	/home user quota	/globalscratch user quota	/nqs size	/scratch/tmp
Brasdor	47 Gb	238 Gb	14 Tb	20 Gb - 330 Gb
Courtenay	--	--	2.0 Tb	100 Gb
Fundy	47 Gb	238 Gb	14 Tb	35 Gb - 100 Gb
Glooscap	13 Gb	48 Gb	2.8 Tb	0 Gb - 15 Gb
Mahone	13 Gb	238 Gb	14 Tb	50 Gb - 100 Gb
Placentia	13 Gb	238 Gb	14 Tb	0 Gb - 140 Gb

Cluster-local

Node-local

Applications

https://wiki.ace-net.ca/index.php/Available_Software

<u>Molecular</u>	<u>Math&CS</u>	<u>Engineering</u>	<u>Earth &Ocean</u>	<u>Bio</u>
Gaussian	spin	ANSYS		mpiBlast
Gromacs	GAP	Fluent/	CDO	MrBayes
CPMD	DiVinE	Gambit	GMT	IM
Abinit	Macaulay-2	CFX	MM5	Migrate-n
NAMD	<i>& more</i>	LS-DYNA		PhyML
WebMO		elmer-fem	<u>Astro</u>	PLINK
VMD			StarLab	
<i>& more</i>			IRAF	

Software requests to:

support@ace-net.ca

Software Development

https://wiki.ace-net.ca/index.php/Available_Software

<u>Languages</u>	<u>Suites</u>	<u>Parallel</u> <u>APIs</u>	<u>Libraries</u>	<u>Tools</u>
C/C++	PGI		LAPACK	TotalView
Fortran	SunStudio	Open MPI	FFTW	dbx, gdb
Perl	Gnu	OpenMP	GSL	Valgrind
Python		<i>& more</i>	ACML	make
Java			netCDF	cvs, svn
<i>& more</i>			HDF	git
			Boost	<i>& more</i>
			<i>& more</i>	

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Data Analysis



https://wiki.ace-net.ca/index.php/Available_Software

R

Rmpi

Mathematica

Matlab*

*Distributed Computing Server
Octave

Python

Numpy

SciPy

Sage

pyMPI

pylab/

matplotlib

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Remote Collaboration



Internet videoconferencing - inSORS & Access Grid
Dalhousie - Memorial - Mount Allison - Saint Mary's - Saint Francis Xavier
Acadia - New Brunswick **+ other sites across Canada & the world**

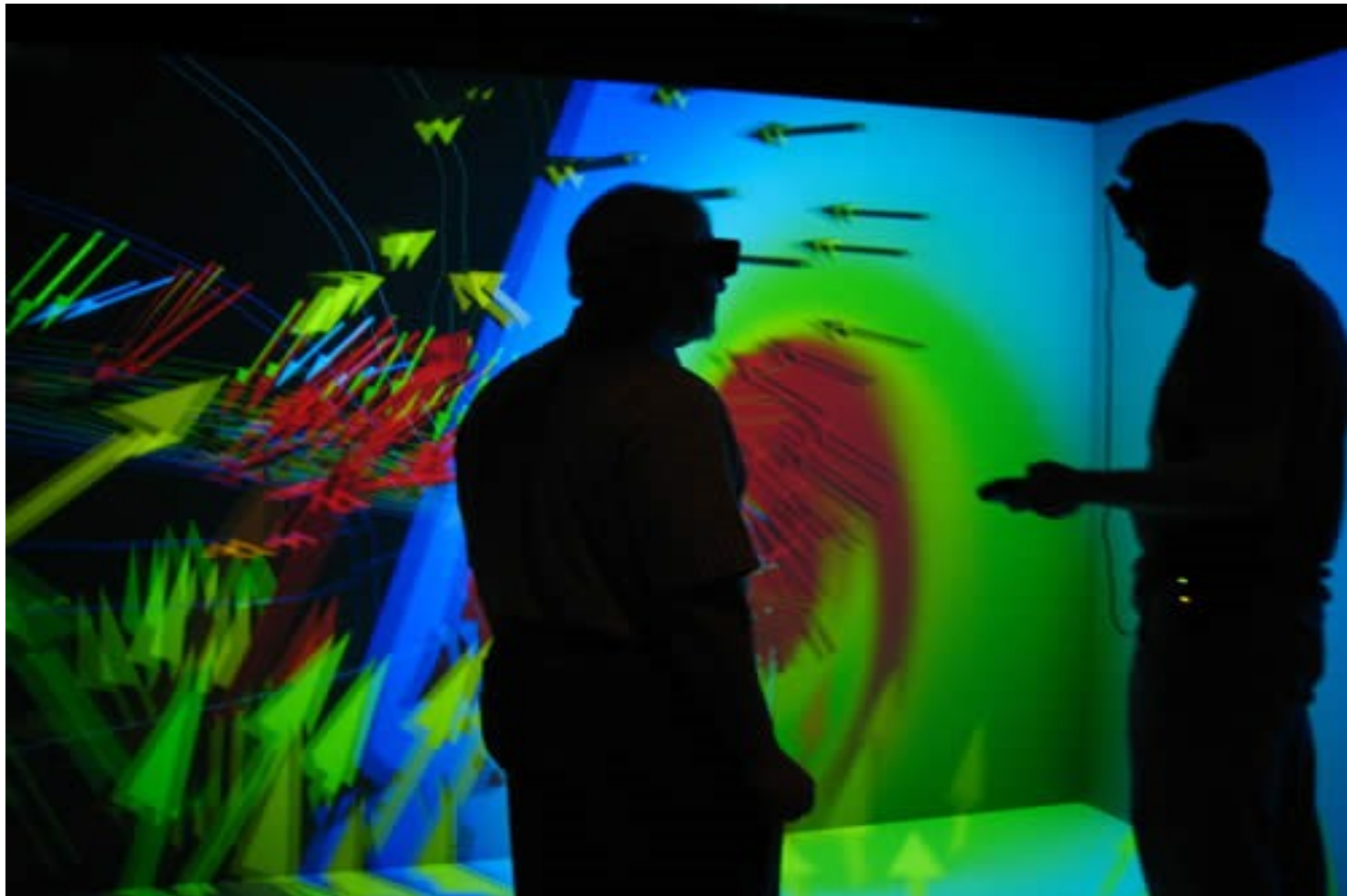


Advanced Visualization



Data cave

3D immersive data exploration facility @ Saint Mary's

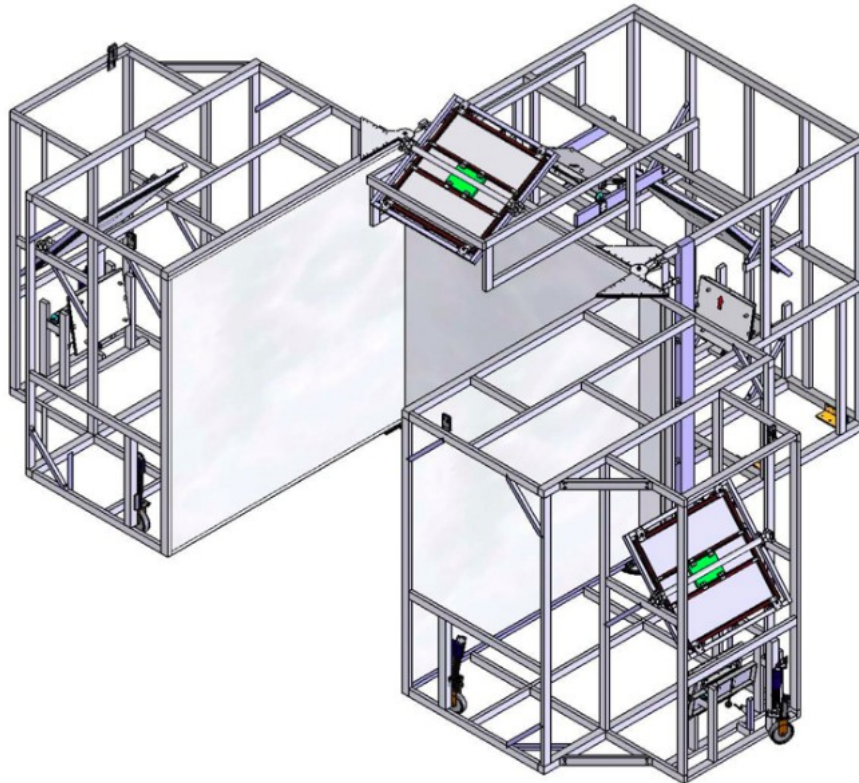


Advanced Visualization



Data cave

3D immersive data exploration facility @ Saint Mary's



Organization and funding

- Consortium of Atlantic Cdn. universities
- Compute Canada is national “umbrella”
- Several \$million from
CFI - ACOA - provinces - others
- ~12 technical staff, + admin
- Research Directorate
Local User Groups ← user direction!
Resource Allocation Committee

Where is ACEnet?



Member institutions



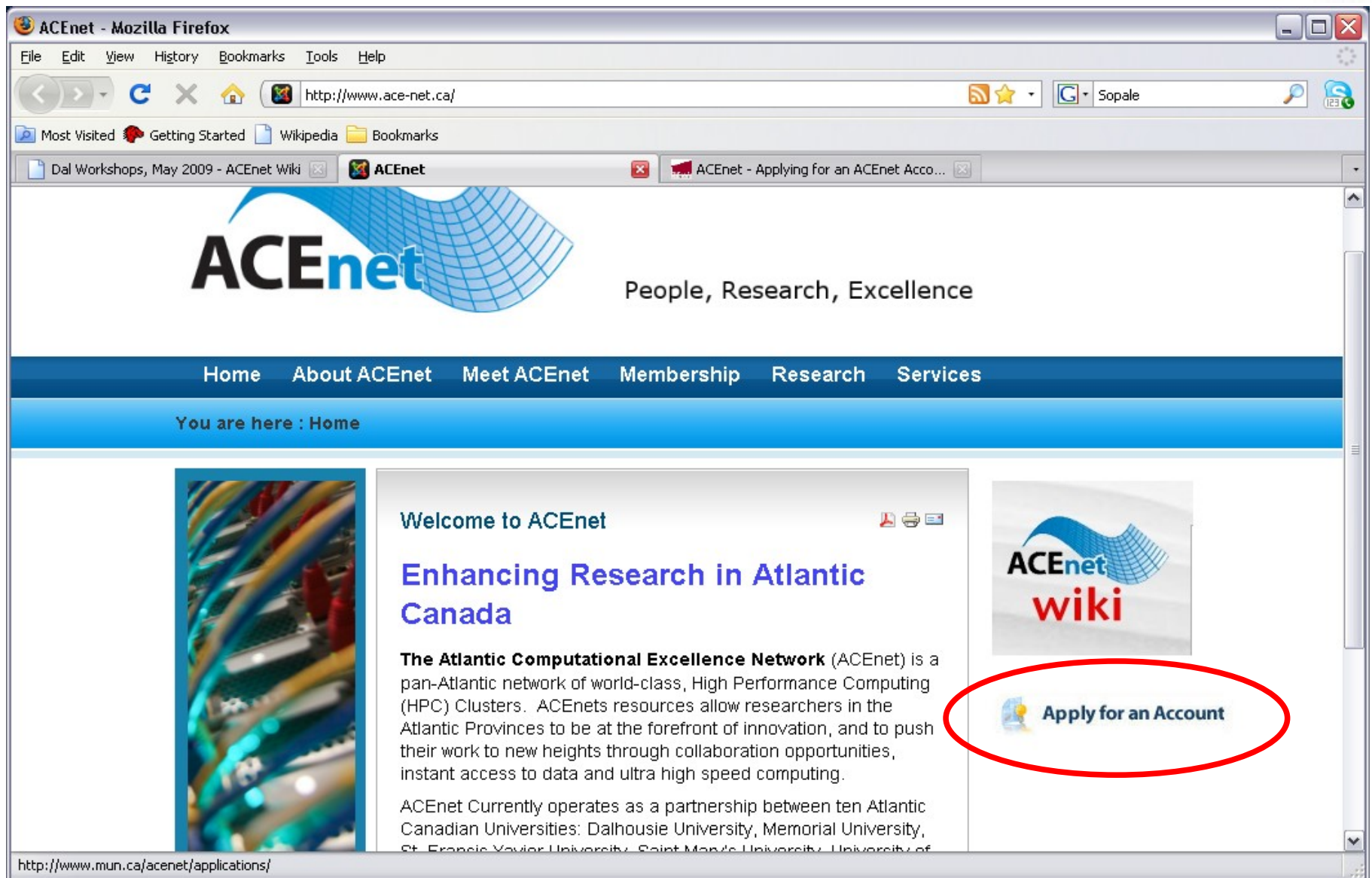
Mount
Allison
UNIVERSITY



CAPE BRETON
UNIVERSITY



Getting an account: <http://www.ace-net.ca/>

A screenshot of a Mozilla Firefox browser window displaying the ACEnet website. The browser's address bar shows the URL "http://www.ace-net.ca/". The website's header includes the ACEnet logo and the tagline "People, Research, Excellence". A navigation menu contains links for Home, About ACEnet, Meet ACEnet, Membership, Research, and Services. Below the menu, a blue banner reads "You are here : Home". The main content area is divided into three columns. The left column features a vertical image of colorful fiber optic cables. The middle column has a "Welcome to ACEnet" heading, followed by the subheading "Enhancing Research in Atlantic Canada" in blue. The text below describes ACEnet as a pan-Atlantic network of world-class High Performance Computing (HPC) Clusters. The right column contains a "wiki" logo and a button labeled "Apply for an Account", which is circled in red. The browser's status bar at the bottom shows the URL "http://www.mun.ca/acenet/applications/".

ACEnet - Applying for an ACEnet Account - Windows Internet Explorer

http://www.mun.ca/acenet/applications/

Google

WebMail :: Inbox

ACEnet - Applying for an ...

Friday, June 08, 2007

People, Research, Excellence

ACEnet

ABOUT US RESEARCH **USERS** NEWS CONTACT US

- [Categories of Accounts](#)
- [Project Account Application](#)
- [User Account Application](#)

Categories of Accounts

ACEnet issues two types of accounts, **Primary Project Accounts** and **User Accounts**. Primary Project Accounts are held by the Research Project Leader who is typically the principal investigator for a project or the head of a research group. A project Leader is responsible for all other user accounts associated with the project.

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[Project Account Application](#)

User accounts will only be issued on the authority of the Research Project Leader for a particular project.

[User Account Application](#)

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http://www.ace-net.ca/contactus.html

Internet 100%

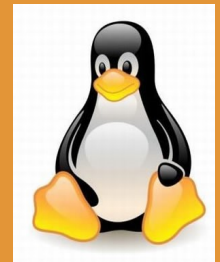
What's the cost?

- No money. It's paid for.
- Tell us about your publications
 - once a year in conjunction with acct renewal
- Share the resource
 - about 250 projects
 - about 1000 user accounts
 - jobs must use dynamic resource manager
 - Sun Grid Engine

How Do I get Connected?



- Logging in
 - ssh *Secure Shell*
- Moving Data Around
 - sftp *Secure File Transfer Protocol*
 - scp *Secure Copy*
- both provided with **Mac OS X** or **Linux**
 - ssh -X username@brasdor.ace-net.ca
 - sftp username@placentia.ace-net.ca
- **Windows?** Use PuTTY and WinSCP
just google for them



Where to go for help

- ACEnet User Guide
 - http://wiki.ace-net.ca/index.php/User_Guide
- ACEnet User Wiki
 - <http://wiki.ace-net.ca>
- Research/user support personnel
 - email: support@ace-net.ca

What cluster should I use?



- Doesn't matter,
ONE ACCOUNT IS GOOD EVERYWHERE
- Multiple serial jobs? Brasdor best
- Shared memory parallel? Fundy best
- MPI? Mahone best, Placentia, Glooscap, Fundy good
- Some software has only local license
 - e.g. Matlab, Gaussian only at Placentia
 - See <http://wiki.ace-net.ca> for availability, or
contact your Computational Research Consultant